Digital SAT Worked Examples — Every Tip in Action

A Comprehensive Step-by-Step Guide to Mastering Every Question Type

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This guide transforms every quick-review tip into a **fully worked, step-by-step example** so you can see exactly how each concept plays out on real SAT-style questions. Every problem is original, every step is shown, and every common trap is flagged. Work through each example carefully, then test yourself with the "Your Turn" problems at the end of each section.

How to Use This Guide

1. Read the **RULE** box at the top of each section — this is the core concept. **2.** Work through each **Example** step-by-step. Cover the solution and try it yourself first. **3.** Pay close attention to **Common Mistake** warnings — these are the exact traps the SAT sets. **4.** Read the **Why This Matters** note to understand the real-test payoff. **5.** Attempt the **Your Turn** practice problem before checking the answer.

PART 1: READING

&

WRITING — DETAILED EXAMPLES

1A. Grammar — Subject-Verb Agreement

> ■ ***RULE***
>
> *When a long phrase sits between the subject and verb, mentally cross it out. The verb must agree with the TRUE subject, not the nearest noun.*

Example 1 (Easy)

Passage: "The discovery of several ancient artifacts _____ researchers to reconsider the timeline of human settlement in the region."

A) have prompted	B) are prompting	C) has prompted	D) prompt

Step 1: Find the subject. → **"The discovery"** — singular.

Step 2: Cross out the intervening phrase: *"of several ancient artifacts"* — this is a prepositional phrase modifying "discovery." It does **NOT** change the subject.

Step 3: Read without the phrase: "The discovery _____ researchers..."

Step 4: Singular subject needs singular verb → **"has prompted" (C)**.

> ⚠️ **Common Mistake**
>
> Students pick (A) "have prompted" because *"artifacts"* (plural) is right next to the blank. But "artifacts" is the object of the preposition "of," not the subject.

✓ **Why This Matters**

This exact pattern appears on every SAT. The test deliberately places plural nouns near the blank to trick you. Learning to identify the true subject instantly earns you 2–3 free points.

Example 2 (Hard)

Passage: "The committee's analysis of the environmental reports, along with recommendations from three independent consultants, _____ that the proposed construction site poses minimal risk to local wildlife."

A) suggest	B) have suggested	C) suggests	D) are suggesting

Step 1: Find the subject. → **"The committee's analysis"** — singular.

Step 2: Cross out ALL intervening material: *"of the environmental reports"* (prepositional phrase) **AND** *"along with recommendations from three independent consultants"* (parenthetical addition).

Step 3: KEY RULE — *"along with," "together with," "as well as,"* and *"in addition to"* do **NOT** make a subject plural. Only **"and"** does.

Step 4: Read without extras: "The analysis _____ that..."

Step 5: Singular subject → singular verb → **"suggests" (C)**.

⚠️ **Common Mistake**

"Along with recommendations" makes it SEEM like there are two

subjects. But grammatically, "along with" is not the same as "and." Only "and" creates a compound subject.

✎ **Your Turn**

"The collection of handwritten letters from the 18th century, which were recently donated by a private collector, _____ on display at the city museum through December."

A) are **B)** is **C)** were **D)** have been

Answer: B — "collection" is singular; "letters" and "collector" are inside prepositional phrases and a nonrestrictive clause.

1B. Grammar — Colon and Semicolon Rules

RULE

Both colons and semicolons require an independent clause (a complete sentence) BEFORE them. Colon = introduces what follows (a list, explanation, or elaboration). Semicolon = joins two complete sentences that are closely related.

Example 1 — Colon (Easy)

Passage: "The research team identified three factors contributing to coral bleaching_____ rising ocean temperatures, increased UV radiation, and chemical runoff from agricultural sites."

A) , which are:	**B) :**	**C) ;**	**D) , including**

Step 1: Check what comes **before** the blank. "The research team identified three factors contributing to coral bleaching" — this is a complete sentence. ✓

Step 2: Check what comes **after**. A list of three items. A colon is perfect for introducing a list after a complete sentence.

Step 3: Check each option:

- (A) ", which are:" — awkward and redundant
- (B) ":" — clean, correct; independent clause before + list after
- (C) ";" — semicolons join two independent clauses, but a list is not a clause
- (D) ", including" — grammatically okay but changes the sentence structure; the original uses "three factors" which sets up a colon naturally

Step 4: Answer: **(B)**

Example 2 — Semicolon (Medium)

Passage: "Dr. Patel's initial hypothesis predicted a linear decline in soil nutrients_____ however, the data revealed an unexpected plateau during the third year of the study."

A) ,	**B) :**	**C) ;**	**D) —**

Step 1: Check before the blank. "Dr. Patel's initial hypothesis predicted a linear decline in soil nutrients" — complete sentence. ✓

Step 2: Check after the blank. "however, the data revealed an unexpected plateau..." — also a complete sentence (starting with a conjunctive adverb). ✓

Step 3: Two independent clauses + "however" in between = **semicolon**.

Step 4: KEY RULE — *"however," "therefore," "moreover," "nevertheless,"* and *"furthermore"* are **NOT** conjunctions. You cannot use just a comma before them. You need a semicolon (or a period).

Step 5: Answer: **(C)**

⚠️ **Common Mistake**

Choosing (A) creates a **comma splice** — two complete sentences joined by only a comma. This is ALWAYS wrong on the SAT when "however" follows.

Example 3 — Colon for Elaboration (Hard)

Passage: "The playwright's latest work represents a significant departure from her earlier style_____ whereas her previous plays relied on naturalistic dialogue, this production employs a fragmented, nonlinear narrative structure."

A) :	**B) ;**	**C) ,**	**D) . And**

Step 1: Before the blank: "The playwright's latest work represents a significant departure from her earlier style" — complete sentence. ✓

Step 2: After the blank: "whereas her previous plays relied on..." — this **EXPLAINS** what the "departure" is. It elaborates on the claim.

Step 3: A colon can introduce an **explanation or elaboration**, not just a list.

Step 4: A semicolon (B) could also work since both sides are independent clauses, but (A) is stronger because the second clause specifically *explains* the first.

Step 5: Answer: **(A)**

✓ **Why This Matters**

Students often think colons are only for lists. On the harder Module 2, colons introduce explanations. Recognizing this pattern is worth 1–2 extra points.

✎ **Your Turn**

"The museum's newest exhibit explores a single, powerful theme_____ the resilience of communities in the face of environmental disaster."

A) , **B)** ; **C)** : **D)** — being

Answer: C — independent clause before the blank, and what follows is an elaboration/specification of the "theme."

1C. Grammar — Verb Tense and Mood

> ■ ***RULE***
>
> *Subjunctive mood: after verbs like "insists," "demands," "recommends," "requires," "suggests," use the BASE form of the verb (no -s, no -ed).*
>
> *Past perfect: use "had + past participle" when one past event happened BEFORE another past event.*

Example 1 — Subjunctive (Medium)

Passage: "The department chair insists that every graduate student _____ a revised thesis outline before the end of the semester."

A) submits	B) submit	C) will submit	D) would submit

Step 1: Identify the trigger word → **"insists that"** — this demands the subjunctive mood.

Step 2: Subjunctive rule: after "insists/demands/recommends/requires that," use the **BASE form** of the verb.

Step 3: Base form of "submit" = **"submit"** (no -s, no tense change).

Step 4: Answer: **(B)**

> ⚠️ **Common Mistake**
>
> (A) "submits" sounds natural in everyday speech ("every student submits"), but after "insists that," the SAT requires the subjunctive base

form.

Example 2 — Past Perfect (Medium)

Passage: "By the time emergency crews arrived at the riverbank, floodwaters _____ several homes and displaced dozens of families."

A) destroyed	B) have destroyed	C) had destroyed	D) were destroying

Step 1: Identify the time sequence. Two past events: (1) floodwaters destroyed homes, (2) emergency crews arrived.

Step 2: Which happened first? The destruction happened **BEFORE** the crews arrived.

Step 3: When one past action comes before another past action → past perfect (**"had + past participle"**).

Step 4: "had destroyed" correctly shows that the destruction was already complete when the crews arrived.

Step 5: Answer: **(C)**

Example 3 — Contrary-to-Fact Conditional (Hard)

Passage: "If the expedition team _____ additional funding before the winter season, they might have reached the summit before the storms made the final ascent impossible."

A) secured	B) had secured	C) has secured	D) would have secured

Step 1: This is a conditional sentence about something that **DID NOT HAPPEN** (they didn't get funding, they didn't reach the summit).

Step 2: Contrary-to-fact past conditional structure: *"If [subject] **HAD** [past participle], [subject] might/would/could have [past participle]."*

Step 3: The "might have reached" in the second clause confirms this is a past contrary-to-fact conditional.

Step 4: The "if" clause needs **"had secured."**

Step 5: Answer: **(B)**

✎ **Your Turn**

"The director recommends that the new policy _____ implemented gradually over the next fiscal year to minimize disruption."

A) is **B)** be **C)** will be **D)** should be

Answer: B — "recommends that" triggers subjunctive; base form of "to be" = "be."

1D. Grammar — Comma Rules (Nonrestrictive Clauses)

■ RULE

If you can remove the phrase and the sentence still makes complete sense, it's nonrestrictive — surround it with commas. If removing it changes the meaning or makes the sentence unclear, it's restrictive —

no commas.

Example 1 (Medium)

Passage: "Dr. Elena Vasquez_____ who has studied marine ecosystems for over two decades_____ recently published findings suggesting that deep-sea coral reefs may be more resilient to temperature changes than previously believed."

A)	**, who has studied marine ecosystems for over two decades,**
B)	who has studied marine ecosystems for over two decades
C)	, who has studied marine ecosystems for over two decades
D)	who has studied marine ecosystems for over two decades,

Step 1: Identify the clause: *"who has studied marine ecosystems for over two decades."*

Step 2: Removal test: "Dr. Elena Vasquez recently published findings..." — still makes complete sense. We know exactly who Dr. Vasquez is from her name.

Step 3: Since removing the clause doesn't change the core meaning → it's **nonrestrictive** → surround with commas on **BOTH sides**.

Step 4: Answer: **(A)** — commas before AND after.

⚠️ **Common Mistake**

Students often put a comma only on one side — (C) or (D).

Nonrestrictive clauses need **MATCHING commas**, like parentheses.

Example 2 — Paired Em Dashes (Hard)

Passage: "The cellist's performance — a breathtaking interpretation of Bach's Suite No. 1_____ earned a standing ovation from the audience."

A) ,	**B) ;**	**C) —**	**D) :**

Step 1: Notice the opening em dash before *"a breathtaking interpretation."*

Step 2: "a breathtaking interpretation of Bach's Suite No. 1" is an inserted aside — nonrestrictive.

Step 3: Em dashes work like commas for nonrestrictive elements, but they come in **PAIRS**. If one dash opens the aside, another must close it.

Step 4: Answer: **(C)** — the closing em dash matches the opening one.

Key Rule

Dashes, like commas and parentheses, must come in **matched pairs** when used for insertions mid-sentence.

✎ **Your Turn**

"The novel_____ which was translated into 14 languages_____ explores themes of displacement and cultural identity through the eyes of three

generations of immigrants."

A) , which was translated into 14 languages, **B)** which was translated into 14 languages

C) , which was translated into 14 languages **D)** which was translated into 14 languages,

Answer: A — "which was translated into 14 languages" is nonrestrictive and needs commas on both sides.

1E. Words in Context

> ■ ***RULE***
> *Never pick the most common dictionary definition. Read the full sentence, substitute each choice into the blank, and pick the one that preserves the author's intended meaning in context.*

Example 1 (Easy)

Passage: "The city council's decision to reduce funding for public parks was met with considerable opposition, as many residents argued that green spaces are essential for **mitigating** the stresses of urban living."

As used in the passage, **"mitigating"** most nearly means:

A) eliminating	**B) understanding**	**C) lessening**	**D) identifying**

Step 1: Read the sentence. Green spaces help with "mitigating the stresses of urban living."

Step 2: Context clues: parks are "essential" for dealing with stress — they don't eliminate stress entirely, but they help reduce it.

Step 3: Substitute each choice:

- (A) "eliminating the stresses" — too strong; parks don't remove all stress
- (B) "understanding the stresses" — doesn't fit; the passage is about relief, not comprehension
- (C) **"lessening the stresses" — fits perfectly; parks reduce stress**
- (D) "identifying the stresses" — wrong direction; not about finding stress

Step 4: Answer: **(C)**

Example 2 (Hard)

Passage: "While the ambassador publicly endorsed the proposed trade agreement, several members of the diplomatic corps noted that she could barely **countenance** the concessions her government had been forced to accept during the final round of negotiations."

As used in the passage, **"countenance"** most nearly means:

A) tolerate	B) acknowledge	C) envision	D) display

Step 1: Read carefully. The ambassador "publicly endorsed" the agreement **BUT** "could barely countenance the concessions."

Step 2: The word "barely" + the contrast with "publicly endorsed" tells us she's struggling with something internally.

Step 3: Substitute:

- (A) **"barely tolerate the concessions" — fits: she accepted them publicly but could hardly stand them**
- (B) "barely acknowledge" — possible, but she DID publicly endorse them, so she acknowledged them
- (C) "barely envision" — doesn't match; the concessions already happened
- (D) "barely display" — doesn't make grammatical sense here

Step 4: Answer: **(A)**

✓ **Why This Matters**

"Countenance" as a verb meaning "tolerate" is exactly the kind of high-register word that appears on Module 2. The test loves words with uncommon secondary meanings.

✎ **Your Turn**

"The researcher's findings were preliminary, and she cautioned that the data should not be used to draw definitive conclusions, as the sample size was too **modest** to warrant such confidence."

As used in the passage, "modest" most nearly means:

A) humble **B)** small **C)** decent **D)** shy

Answer: B — in context, "modest" describes the sample size being too small to support strong conclusions, not the common meaning of "humble/shy."

1F. Transitions

RULE

*There are only 5 transition types: **CAUSE** (therefore, consequently, as a result), **CONTRAST** (however, nevertheless, on the other hand), **ADDITION** (moreover, furthermore, in addition), **EXAMPLE** (for instance, for example), **SUMMARY** (in other words, in short). Read the sentence BEFORE and AFTER the blank — the relationship between them is your answer.*

Example 1 — Cause/Effect (Easy)

Passage: "The region experienced unprecedented rainfall throughout the spring, with some areas receiving more than double their historical average. _____, several major rivers exceeded their flood stages, forcing the evacuation of thousands of residents."

A) Nevertheless	B) For example	C) As a result	D) In contrast

Step 1: Sentence before: unprecedented rainfall (CAUSE).

Step 2: Sentence after: rivers flooded, evacuations happened (EFFECT).

Step 3: Cause → Effect = **"As a result."**

Step 4: Check others:

- (A) "Nevertheless" = contrast — but flooding is a logical consequence of rain, not a surprise
- (B) "For example" = the flooding isn't an example of rain; it's a result of it
- (D) "In contrast" = no contradiction here

Step 5: Answer: **(C)**

Example 2 — Contrast (Medium)

Passage: "Initial laboratory trials of the compound showed remarkable promise in reducing tumor growth rates by nearly 40 percent. _____, subsequent clinical trials involving human participants yielded far more modest results, with reductions averaging only 12 percent."

A) Moreover	B) Therefore	C) Similarly	D) However

Step 1: Sentence before: lab trials showed "remarkable promise" — 40% reduction (positive).

Step 2: Sentence after: human trials were "far more modest" — only 12% (disappointing).

Step 3: Positive result → Disappointing result = **CONTRAST**.

Step 4: "However" signals contrast. Answer: **(D)**

⚠️ **Common Mistake**

Students sometimes pick (A) "Moreover" because both sentences discuss trial results. But "moreover" means "in addition to the same point," while here the results **CONTRADICT** each other.

Example 3 — Addition (Hard)

Passage: "The new building's design incorporates passive solar heating, which reduces energy consumption during winter months by an estimated 30 percent. _____, the rooftop garden functions as natural insulation, further decreasing the need for mechanical climate control systems throughout the year."

A) However	B) For instance	C) In other words	D) Moreover

Step 1: Sentence before: passive solar heating reduces energy use (benefit #1).

Step 2: Sentence after: rooftop garden decreases energy needs (benefit #2 — same direction).

Step 3: Adding a second supporting point in the same direction = **ADDITION**.

Step 4: "Moreover" means "in addition, here's another supporting point." Answer: **(D)**

Step 5: Why not (B) "For instance"? The rooftop garden is a **SEPARATE feature**, not an example of passive solar heating.

✎ **Your Turn**

"The orchestra's performance of the symphony's first movement was technically flawless, with every section executing the complex passages with precision. _____, the emotional depth that critics had praised in rehearsals was noticeably absent during the live performance."

A) As a result **B)** Furthermore **C)** Nevertheless **D)** For example

Answer: C — "Nevertheless" signals contrast: technically perfect BUT emotionally flat.

1G. Inference Questions

■ ***RULE***

The answer is ALWAYS supported by specific text. If you can't point to a line in the passage that supports your answer, it's wrong. Eliminate any choice with extreme language ("always," "never," "proves," "all") — the SAT rarely supports absolute claims.

Example 1 (Medium)

Passage: "A 2023 survey of 1,200 commuters in a mid-sized city found that 68% drove alone to work, 18% used public transit, 9% carpooled, and 5% bicycled or walked. When asked about willingness to switch to public

transit, 41% of solo drivers cited unreliable schedules as their primary concern, while 27% mentioned insufficient routes."

Based on the data in the passage, which of the following can most reasonably be inferred?

A) Most commuters in the city prefer driving because public transit is too expensive.
B) Improving schedule reliability would likely have a greater impact on ridership than expanding routes.
C) The city should invest in bicycle infrastructure rather than public transit.
D) All solo drivers would switch to public transit if schedules improved.

Step 1: Evaluate each choice against the passage:

- (A) "too expensive" — the passage **never mentions cost**. ELIMINATE — unsupported.
- (B) "schedule reliability > expanding routes" — 41% cited schedules vs. 27% cited routes. 41% > 27%, so addressing schedules would affect more people. This is **SUPPORTED** by the data.
- (C) "invest in bicycles rather than transit" — only 5% bike/walk; the passage doesn't compare investment priorities. ELIMINATE.
- (D) "**All** solo drivers would switch" — EXTREME LANGUAGE. Only 41% cited schedules. ELIMINATE.

Step 2: Answer: **(B)**

Key Technique

Point to the exact data. 41% > 27% → schedule reliability has a larger impact. This is a reasonable inference from the numbers, not an assumption.

Example 2 — Command of Evidence (Hard)

Passage: "Dr. Kimura's research team hypothesized that bilingual children develop stronger executive function skills — the mental processes involved in planning, attention, and task-switching — than their monolingual peers. The team administered a series of cognitive tests to 200 children aged 8–12, half bilingual and half monolingual. On tests measuring task-switching speed, bilingual children performed 15% faster on average. However, on tests of sustained attention and working memory, the two groups showed no statistically significant difference."

Which finding, if true, would most **weaken** Dr. Kimura's hypothesis?

A) Bilingual children in the study came from families with higher average household incomes.
B) The task-switching test was administered in English, which all participants spoke fluently.
C) A separate study found similar results in a population of bilingual adults.
D) Monolingual children spent more time on extracurricular reading activities.

Step 1: The hypothesis: bilingualism → stronger executive function.

Step 2: The evidence: bilingual kids were faster at task-switching, but

equal on attention and memory.

Step 3: To **WEAKEN**: find an alternative explanation for the task-switching advantage that isn't bilingualism.

Step 4: Evaluate:

- (A) Higher incomes → the advantage might be from resources/education, not bilingualism. This is a **CONFOUNDING VARIABLE**. It weakens the claim that bilingualism caused the improvement. ✓
- (B) Test in English — all spoke English fluently, so no disadvantage. Doesn't weaken.
- (C) Similar results in adults — this would **STRENGTHEN**, not weaken.
- (D) More reading — if anything, this might help monolinguals, but it's not directly related to task-switching.

Step 5: Answer: **(A)** — income is a confounding variable that provides an alternative explanation.

✎ **Your Turn**

"A study of 500 high school students found that those who participated in three or more extracurricular activities had an average GPA of 3.4, while those in fewer than three had an average GPA of 3.1. Students in three or more activities also reported sleeping an average of 45 minutes less per night."

Based on the passage, which inference is best supported?

A) Extracurricular activities directly cause higher GPAs.

B) Students who participate in more activities tend to have higher GPAs, but they may sacrifice sleep.

C) All students should join three or more activities to improve their grades.

D) Sleep deprivation has no effect on academic performance.

Answer: B — the data shows a correlation between activities and GPA, plus a trade-off with sleep. (A) claims causation, (C) uses "all," and (D) contradicts the sleep finding.

PART 2: MATHEMATICS — DETAILED EXAMPLES

2A. Algebra — Systems with No Solution / Infinite Solutions

RULE

For a system of two linear equations: ***ONE SOLUTION*** *= different slopes (lines intersect).* ***NO SOLUTION*** *= same slope, different y-intercept (parallel lines).* ***INFINITE SOLUTIONS*** *= equations are proportional / same line.*

Example 1 — No Solution (Medium)

Problem: For what value of k does the following system have no solution?

$3x + 6y = 12$

$kx + 2y = 7$

Step 1: For no solution, the lines must be parallel (same slope, different intercept).

Step 2: Rewrite both in slope-intercept form ($y = mx + b$):

Equation 1: $6y = -3x + 12 \rightarrow y = -1/2\ x + 2 \rightarrow$ **slope = -1/2**

Equation 2: $2y = -kx + 7 \rightarrow y = -k/2\ x + 7/2 \rightarrow$ **slope = -k/2**

Step 3: Set slopes equal: $-k/2 = -1/2 \rightarrow$ **k = 1**

Step 4: Verify intercepts are different: Eq 1 intercept = 2, Eq 2 intercept = 7/2. Since $2 \neq 7/2$, confirmed — parallel lines, no solution.

Step 5: Answer: **k = 1**

Example 2 — Infinite Solutions (Hard)

Problem: For what value of c does the following system have infinitely many solutions?

$4x + 6y = 14$
$2x + 3y = c$

Step 1: For infinite solutions, the equations must be the **SAME line** (one is a multiple of the other).

Step 2: Compare the equations. Multiply Equation 2 by 2:

$2(2x + 3y) = 2c \rightarrow 4x + 6y = 2c$

Step 3: This must equal Equation 1: $4x + 6y = 14$

Step 4: So $2c = 14 \rightarrow$ **c = 7**

Step 5: Answer: **c = 7**

⚠️ **Common Mistake**

Students confuse no solution and infinite solutions. Remember: **No solution** = same slope, DIFFERENT intercept. **Infinite** = same slope, SAME intercept (identical lines).

✎ **Your Turn**

For what value of *m* does this system have no solution?

5x - 10y = 20

mx - 4y = 9

Answer: m = 2. Eq 1: y = 1/2 x - 2, slope = 1/2. Eq 2: y = m/4 x - 9/4, slope = m/4. Set m/4 = 1/2, so m = 2. Intercepts -2 ≠ -9/4, confirmed parallel.

2B. Algebra — Absolute Value Equations

■ ***RULE***

|expression| = k gives TWO equations: expression = k AND expression = -k. Always check: if k < 0, there is NO solution (absolute value can never be negative).

Example (Medium)

Problem: What is the sum of all solutions to |2x - 5| = 9?

Step 1: Split into two equations:

Case 1: $2x - 5 = 9$

Case 2: $2x - 5 = -9$

Step 2: Solve Case 1:

$2x - 5 = 9 \rightarrow 2x = 14 \rightarrow$ **$x = 7$**

Step 3: Solve Case 2:

$2x - 5 = -9 \rightarrow 2x = -4 \rightarrow$ **$x = -2$**

Step 4: Sum of solutions: $7 + (-2) =$ **5**

Step 5: Answer: **5**

⚠️ **Common Mistake**

Only solving one case and getting $x = 7$ alone. Absolute value ALWAYS creates two cases.

✎ **Your Turn**

What is the product of all solutions to $|3x + 6| = 12$?

Answer: -12. Case 1: $3x + 6 = 12 \rightarrow x = 2$. Case 2: $3x + 6 = -12 \rightarrow x = -6$. Product = $2 \times (-6) = -12$.

2C. Advanced Math — Completing the Square

RULE

To convert $x^2 + bx + c$ to vertex form: $x^2 + bx + c = (x + b/2)^2 - (b/2)^2 + c$. Take half of b, square it, add and subtract.

Example 1 (Hard)

Problem: The function $f(x) = x^2 + 8x + 12$ can be rewritten as $f(x) = (x + a)^2 + b$. What is the value of b?

Step 1: Start with $x^2 + 8x + 12$.

Step 2: Take half of the coefficient of x: 8/2 = **4**.

Step 3: Square it: 4^2 = **16**.

Step 4: Add and subtract 16 inside the expression:

$x^2 + 8x + 16 - 16 + 12$

Step 5: Group the perfect square trinomial:

$(x^2 + 8x + 16) - 16 + 12 = \mathbf{(x + 4)^2 - 4}$

Step 6: So a = 4 and b = -4.

Step 7: Answer: **b = -4**

Visual Check

The vertex is at (-4, -4). Since the coefficient of x^2 is positive (opens up), -4 is the **minimum** value.

Example 2 — Finding the Maximum (Hard)

Problem: A ball is thrown upward. Its height in feet after *t* seconds is $h(t) = -t^2 + 6t + 2$. What is the maximum height the ball reaches?

Step 1: Factor out -1 from the quadratic terms:

$h(t) = -(t^2 - 6t) + 2$

Step 2: Complete the square inside the parentheses:

Half of -6 = -3, squared = 9

$h(t) = -(t^2 - 6t + 9 - 9) + 2$

$h(t) = -((t - 3)^2 - 9) + 2$

$h(t) = -(t - 3)^2 + 9 + 2$

$h(t) = \mathbf{-(t - 3)^2 + 11}$

Step 3: Since the coefficient is negative (parabola opens down), the vertex is the **MAXIMUM**.

Step 4: Maximum height = **11 feet**, occurring at t = 3 seconds.

Step 5: Answer: **11 feet**

Alternative Method

Use the vertex formula: $t = -b/2a = -6/(2 \times (-1)) = 3$. Then $h(3) = -(9) + 18 + 2 = 11$. Same answer!

✎ **Your Turn**

Rewrite $g(x) = x^2 - 10x + 21$ in vertex form. What is the minimum value

of g(x)?

Answer: Minimum = -4. $x^2 - 10x + 25 - 25 + 21 = (x - 5)^2 - 4$. Minimum value = -4.

2D. Advanced Math — Function Composition

> ■ ***RULE***
>
> *f(g(x)): Always work INSIDE OUT. Calculate g(x) first, then plug that result into f.*

Example 1 (Hard)

Problem: If $f(x) = x^3 - 12$ and $g(x) = 2x + 1$, what is the value of f(g(1))?

Step 1: Work inside out. Start with the inner function: g(1).

$g(1) = 2(1) + 1 =$ **3**

Step 2: Now plug that result into f:

$f(3) = 3^3 - 12 = 27 - 12 =$ **15**

Step 3: Answer: **f(g(1)) = 15**

> ⚠️ **Common Mistake**
>
> Computing f(1) first, then g(f(1)). That gives g(f(1)) = g(-11) = -21, which is f(g(1)) backwards. **ALWAYS go inside out.**

Example 2 — Algebraic Composition (Hard)

Problem: If $f(x) = x^2 + 1$ and $g(x) = x - 3$, what is $f(g(x))$?

Step 1: Replace x in f with g(x):

$f(g(x)) = (g(x))^2 + 1$

Step 2: Substitute $g(x) = x - 3$:

$f(g(x)) = (x - 3)^2 + 1$

Step 3: Expand:

$= x^2 - 6x + 9 + 1 = \mathbf{x^2 - 6x + 10}$

Step 4: Answer: $\mathbf{f(g(x)) = x^2 - 6x + 10}$

✎ **Your Turn**

If $f(x) = 3x - 2$ and $g(x) = x^2 + 4$, what is $f(g(2))$?

Answer: 22. $g(2) = 4 + 4 = 8$. $f(8) = 24 - 2 = 22$.

2E. Advanced Math — Exponent Rules

■ ***RULE***

Key rules: (1) $(ab)^n = a^n \times b^n$ (2) $(a^m)^n = a^{mn}$ (3) $a^m / a^n = a^{m-n}$ (4) $a^m \times a^n = a^{m+n}$ (5) $a^0 = 1$ (6) $a^{-n} = 1/a^n$

Example (Hard)

Problem: Simplify: $(4x^2y^3)^2 / (2xy)^2$

Step 1: Expand the numerator using rules (1) and (2):

$(4x^2y^3)^2 = 4^2 \times (x^2)^2 \times (y^3)^2 = \mathbf{16x^4y^6}$

Step 2: Expand the denominator:

$(2xy)^2 = 2^2 \times x^2 \times y^2 = \mathbf{4x^2y^2}$

Step 3: Divide using rule (3):

$16x^4y^6 / (4x^2y^2) = (16/4) \times x^{4-2} \times y^{6-2} = \mathbf{4x^2y^4}$

Step 4: Answer: $\mathbf{4x^2y^4}$

⚠️ **Common Mistake**

Forgetting to apply the exponent to the coefficient. $(4x^2y^3)^2$ — the 4 also gets squared to 16, not left as 4.

✎ **Your Turn**

Simplify: $(3a^3b)^2 / (9a^2b^2)$

Answer: a^4. Numerator: $9a^6b^2$. Denominator: $9a^2b^2$. Divide: $9a^6b^2 / (9a^2b^2) = a^4$.

2F. Data Analysis — Sequential Percent Change

> ■ ***RULE***
>
> *Sequential percent changes are NOT additive. Multiply the multipliers. A p% increase = multiply by (1 + p/100). A p% decrease = multiply by (1 - p/100).*

Example 1 (Hard)

Problem: A store increases the price of a jacket by 20%, then offers a 25% discount on the new price. What is the net percent change from the original price?

Step 1: Start with any convenient original price. Let P = $100.

Step 2: 20% increase: $100 × 1.20 = **$120**

Step 3: 25% decrease on NEW price: $120 × 0.75 = **$90**

Step 4: Net change: ($90 - $100) / $100 = **-10%**

Step 5: Answer: **10% decrease**

Why +20% then -25% ≠ -5%

The 25% discount applies to the HIGHER price ($120), not the original ($100). So the decrease is 25% of $120 = $30, while the increase was only 20% of $100 = $20. Net: $20 up - $30 down = $10 down = 10% decrease.

Formula Method: 1.20 × 0.75 = 0.90 = 90% of original = 10% decrease.

Example 2 (Hard)

Problem: A population increases by 50% in the first decade, then decreases by 30% in the second decade. What is the net percent change over 20 years?

Step 1: Multipliers: 1.50 × 0.70 = **1.05**

Step 2: 1.05 = 105% of original = **5% increase**.

Step 3: Answer: **5% increase** (NOT +20% as simple addition would suggest)

✎ **Your Turn**

A stock drops 40% in Year 1, then rises 50% in Year 2. What is the net change?

Answer: 10% decrease. 0.60 × 1.50 = 0.90 = 10% decrease. Even though +50% > 40%, the 50% gain applies to the reduced value.

2G. Data Analysis — Inclusion-Exclusion Probability

■ RULE

P(A or B) = P(A) + P(B) - P(A and B). You subtract the overlap to avoid double-counting students/items in both groups.

Example (Hard)

Problem: In a school of 200 students, 120 play a sport, 80 are in a music program, and 40 do both. If a student is selected at random, what is the probability that the student plays a sport or is in a music program?

Step 1: Identify values:

P(sport) = 120/200 P(music) = 80/200 P(both) = 40/200

Step 2: Apply inclusion-exclusion:

P(sport OR music) = 120/200 + 80/200 - 40/200

Step 3: Calculate:

= (120 + 80 - 40) / 200 = 160/200 = **4/5** or 0.8

Step 4: Answer: **4/5**

Why Subtract?

Without subtracting, the 40 students who do BOTH get counted once under "sport" and again under "music." Subtracting removes the double-count.

✎ **Your Turn**

In a class of 30 students, 18 study French, 12 study Spanish, and 5 study both. What fraction of students study French or Spanish?

Answer: 5/6. (18 + 12 - 5) / 30 = 25/30 = 5/6.

2H. Geometry — Sector Area

RULE

Sector area = (central angle / 360) × π × r^2. A sector is just a "slice" of a circle — the fraction of the angle out of 360 degrees times the full area.

Example (Hard)

Problem: A circle has a radius of 9. What is the area of a sector with a central angle of 60 degrees?

Step 1: Write the formula: Area = (θ/360) × π × r^2

Step 2: Substitute: Area = (60/360) × π × 9^2

Step 3: Simplify the fraction: 60/360 = **1/6**

Step 4: Calculate: (1/6) × π × 81 = 81π/6 = **13.5π**

Step 5: Answer: **13.5π** (or approximately 42.4)

Visual Thinking

60 degrees is 1/6 of a full circle (360/60 = 6 equal slices). So the sector area is 1/6 of the total area. Total area = 81π, so sector = 81π/6 = 13.5π.

✎ **Your Turn**

A pizza with radius 12 inches is cut into 8 equal slices. What is the area of one slice?

Answer: 18π square inches. Each slice has central angle 360/8 = 45 degrees. Area = (45/360) × π × 144 = (1/8) × 144π = 18π.

2I. Geometry — Special Right Triangles

■ ***RULE***

30-60-90 triangle: *sides are x, x√3, 2x (shortest side opposite 30°, longest = hypotenuse opposite 90°).* ***45-45-90 triangle:*** *sides are x, x, x√2 (two equal legs, hypotenuse = leg × √2).*

Example 1 — 30-60-90 (Hard)

Problem: In triangle DEF, angle D = 30°, angle E = 90°, and EF = 10. What is the length of DF?

Step 1: Identify the triangle type: angles 30-60-90.

Step 2: Label the sides:

- Angle E = 90° → DF is the **hypotenuse** (opposite the right angle)
- Angle D = 30° → EF is opposite the 30° angle = **shortest side = x**
- Angle F = 60° → DE is opposite the 60° angle = $x\sqrt{3}$

Step 3: EF = x = 10, so:

DF (hypotenuse) = 2x = 2(10) = **20**

Step 4: Answer: **DF = 20**

Example 2 — 45-45-90 (Medium)

Problem: A square has a diagonal of length $8\sqrt{2}$. What is the side length of the square?

Step 1: A square's diagonal creates two 45-45-90 triangles.

Step 2: In a 45-45-90: hypotenuse = leg × $\sqrt{2}$.

Step 3: Diagonal = hypotenuse = $8\sqrt{2}$. So: leg × $\sqrt{2}$ = $8\sqrt{2}$.

Step 4: Divide both sides by $\sqrt{2}$: leg = **8**.

Step 5: Answer: side length = **8**

✎ **Your Turn**

In a 30-60-90 triangle, the side opposite the 60° angle is $5\sqrt{3}$. What is the hypotenuse?

Answer: 10. Side opposite 60° = $x\sqrt{3}$ = $5\sqrt{3}$, so x = 5. Hypotenuse = 2x = 10.

2J. Geometry — Composite Solid Volume

> ■ ***RULE***
>
> *Break the shape into basic shapes. Calculate each volume separately, then add or subtract. Key formulas:* ***Cylinder*** *= $\pi r^2 h$.* ***Cone*** *= $(1/3)\pi r^2 h$.* ***Sphere*** *= $(4/3)\pi r^3$.* ***Hemisphere*** *= $(2/3)\pi r^3$.*

Example (Hard)

Problem: A cylinder with radius 4 and height 10 has a cone removed from its interior. The cone has the same radius and height as the cylinder. What is the volume of the remaining solid?

Step 1: Volume of cylinder = $\pi \times r^2 \times h = \pi \times 16 \times 10 =$ **160π**

Step 2: Volume of cone = $(1/3) \times \pi \times r^2 \times h = (1/3) \times \pi \times 16 \times 10 =$ **$160\pi/3$**

Step 3: Remaining volume = Cylinder - Cone:

$160\pi - 160\pi/3$

Step 4: Find common denominator:

$480\pi/3 - 160\pi/3 =$ **$320\pi/3$**

Step 5: Answer: **$320\pi/3$** (approximately 335.1)

Shortcut

Cylinder minus cone of same dimensions = $(2/3) \times \pi \times r^2 \times h$. This works because the cone is exactly 1/3 of the cylinder.

✎ **Your Turn**

A hemisphere with radius 6 sits on top of a cylinder with radius 6 and height 8. What is the total volume?

Answer: 432π. Cylinder = π(36)(8) = 288π. Hemisphere = (2/3)π(216) = 144π. Total = 288π + 144π = 432π.

Quick Reference — All Rules at a Glance

Section	Core Rule	Watch For
1A. Subject-Verb	Cross out prepositional phrases; verb agrees with true subject	"along with" ≠ "and"
1B. Colons/Semicolons	Both need independent clause before; colon introduces, semicolon joins	Comma splices with "however"
1C. Tense & Mood	Subjunctive = base form; past perfect = "had + pp" for earlier event	"insists/demands that" triggers
1D. Commas	Nonrestrictive = removable = commas on BOTH sides	Matched pairs (commas, dashes, parens)
1E. Words in Context	Substitute each choice; pick the one matching the author's intent	Secondary word meanings
1F. Transitions	5 types: cause, contrast, addition, example, summary	"Moreover" ≠ contrast
1G. Inference	Must be supported by specific text; eliminate extreme language	Correlation ≠ causation

Section	Core Rule	Watch For
2A. Systems	No solution = parallel; infinite = same line	Same slope ≠ same intercept
2B. Absolute Value	|expr| = k gives TWO cases; k < 0 = no solution	Don't forget the negative case
2C. Completing Square	Half of b, square it, add and subtract	Factor out leading coefficient first
2D. Composition	f(g(x)) = inside out; compute g first	Don't reverse the order
2E. Exponents	Apply exponent to coefficients AND variables	Don't forget the coefficient
2F. Percent Change	Multiply multipliers; not additive	Each % applies to the NEW value
2G. Inclusion-Exclusion	P(A or B) = P(A) + P(B) - P(A and B)	Subtract the overlap
2H. Sector Area	$(\theta/360) \times \pi r^2$	Fraction of the full circle
2I. Special Triangles	30-60-90: x, x√3, 2x | 45-45-90: x, x, x√2	Match sides to correct angles
2J. Composite Volume	Break into parts, add or subtract	Cone = 1/3 cylinder

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Digital SAT Worked Examples — Every Tip in Action

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Work every example. Flag every common mistake. Own every rule. Your score will follow.